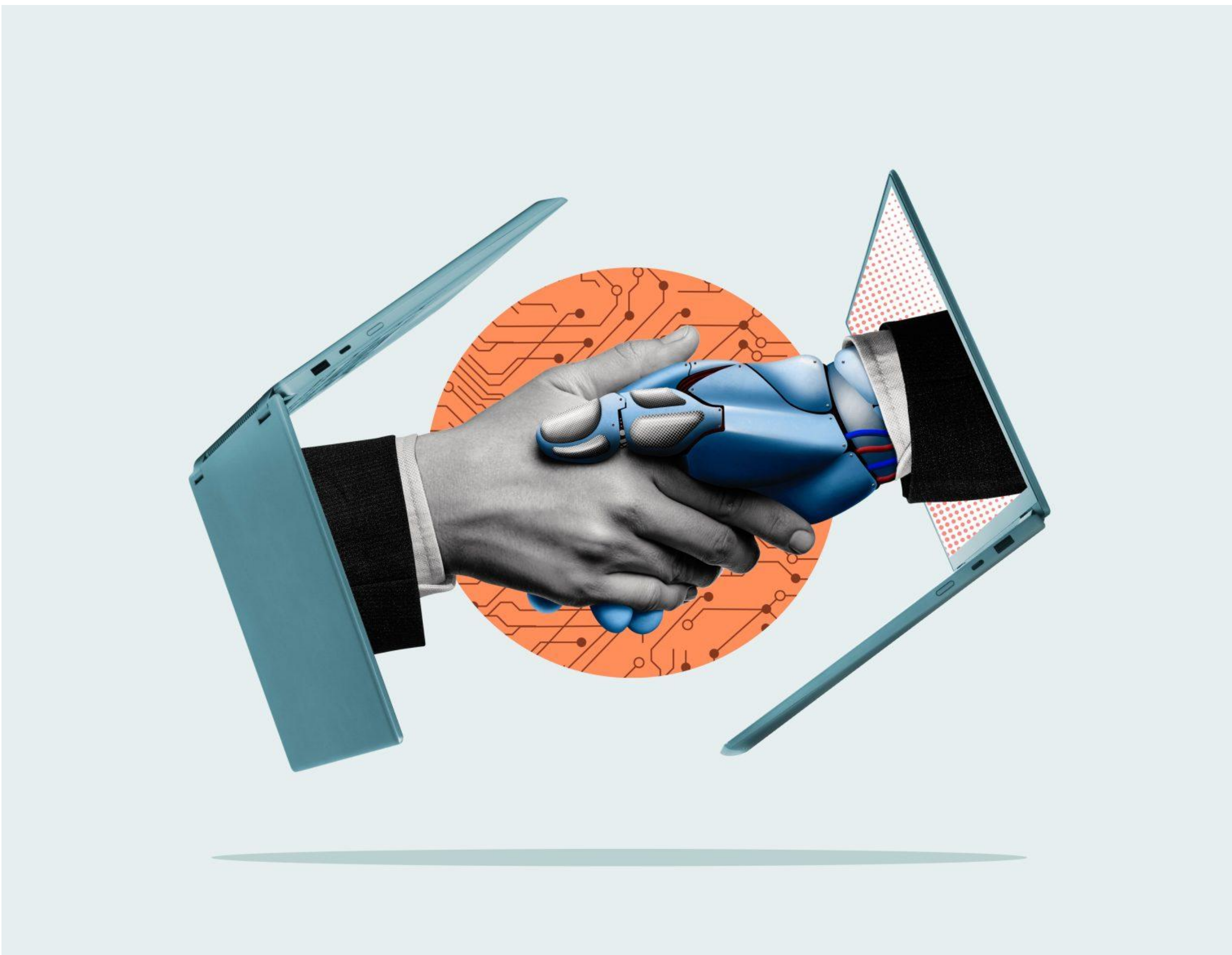
Atlas AI:
Mapping AI and The Global Inequality Gap



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Date of Submission: 5 July 2026

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I. Introduction

Artificial Intelligence (AI) is spreading faster than almost any technology before it, generative AI alone reached three in five U.S. adults within three years of release (CCIA, 2025). Many see this speed as a sign that AI will lift up regions previously left behind. But this promise raises an urgent question: is AI actually closing the global inequality gap, or is it quietly making it wider?

To build this project, our team brings together seven students from seven different academic backgrounds, spanning Data and Society (Aashritha Shrinath), Politics and Technology (Anna Marie Fischer), Computer Science (Eglantine Cruz), Industrial and Complex Systems Engineering (Guillaume Duro Hebert), Offshore Engineering (Rimsha Iftikhar), Civil Engineering (Sandra Martinez Mateos), and Entrepreneurship and Digital Transformation (Ssu Chi Huang). This mix of engineering, policy, data, and business perspectives allowed us to approach the problem from angles a single discipline could not have covered alone.

Today, the creation and control of AI is concentrated among a small number of countries and companies, most of them in the Global North. Just 32 countries host AI-specialized data centers, and Africa and Latin America together account for only 3% of global AI compute capacity, despite representing a large share of the world's population and data (Signé, 2026). Countries in the Global South often supply the raw materials and data that fuel this technology, yet rarely capture its financial benefits or shape how it is governed. Instead, they bear many of its costs, from environmental damage to labor displacement, revealing a deeply unequal system in which a few powerful actors gain while many societies absorb the risks.

To make this imbalance visible, our team built Atlas AI, an interactive website that maps the social, economic, and environmental effects of AI across 35 countries. Data on this topic tends to be scattered and hard to access, especially for the students, researchers, and policymakers who need it most. Atlas AI brings this information together in one place, turning hidden inequalities into something visible and usable, in the hope of guiding AI development toward a fairer, more accountable future.

II. Problem Analysis

Problem Definition

Our project builds on a challenge set by the German Forum for Ethical Machine Decision Making (EME), a nonprofit association promoting the ethical use of AI and machine learning, in collaboration with Young AI Leaders (YAIL). The challenge centers on one critical question:

Is AI truly reducing global inequality, or is it amplifying it?

To properly answer this question, our group was asked to examine AI's global impact across three dimensions. The first is data extraction, where data from the Global South is used to train models that primarily benefit the Global North. The second is unequal access, the gap in infrastructure and connectivity needed to deploy or use advanced AI effectively. The third is labor dynamics, the tension between job displacement and the new economic opportunities automation creates. Our ultimate objective was to identify where the benefits and costs of AI are distributed globally and to visualize these inequalities clearly on a map wherever data was available. By creating this visual prototype, our goal was to offer a clear overview of AI costs versus benefits to better understand what a fair, responsible AI deployment framework should look like.

The Data Gap

Building Atlas AI presented significant challenges during data collection. Although our goal was to map out global AI indicators across all 35 selected nations, we encountered major obstacles that conditioned our workflow and final dataset. The availability of information is unequal and decentralized, while some countries share large amounts of open data, making them rather transparent, others lack this infrastructure. In many cases, the data we needed simply does not exist yet. This lack of transparency and data availability forced us to face empty gaps that could not be filled, meaning certain indicators remain blank for specific countries. A prime example of this was our research into Internet Access (within the Accessibility of AI dimension). Rather than finding one dataset that covered all 35 countries, we relied mainly on Our World in Data and DataReportal's Digital reports, but DataReportal publishes a separate report for each individual country. This meant compiling more than 20 country-specific reports one by one rather than pulling from a single source, which made the process considerably more time-consuming than expected.

These challenges affected Atlas AI in two main ways. First, cleaning and cross-checking data from so many different sources took much longer than expected, which delayed how quickly we could add or update information for certain countries. Second, for a few indicators, so few countries had usable data that including them would have meant showing results for only two or three countries out of 35. In those cases, we decided to leave the indicator out of Atlas AI entirely rather than present a map that was mostly empty. Ultimately, we could only include countries for which sufficient data existed, which meant unintentionally excluding many nations, particularly across parts of the Global South, simply because the information was not accessible or did not exist in a usable form. This in itself reflects the core message of our project: inequality in AI doesn't only exist in its use, but also in how countries measure and share their own data.

Barriers to a Solution

Although the idea of AI dates back to the 1950s, large-scale public use of AI is quite recent. It was not until the release of ChatGPT at the end of 2022 that people began using advanced AI systems and large language models frequently and extensively. AI suddenly became part of everyday life for hundreds of millions of users in just a few months (CCIA, 2025). This rapid adoption means that the social and ecological consequences of AI, such as unequal access, rising energy and water demand, and labor exploitation, have only become visible at scale since around 2023. Researchers and policymakers have relatively little time to build a comprehensive evidence base, let alone coordinated solutions.

At the same time, AI and sustainability remain two unrelated topics. AI metrics are scattered across technical research, corporate disclosures, and specialized indexes, while environmental and social data are held in separate systems and sectoral reports. Very few actors try to connect these domains systematically. As our own data search showed, information relevant to AI inequality is often incomplete, inconsistent across countries, and spread across many institutions, ranging from the World Bank, ITU, IEA, OECD, national labs, and industry groups, to investigative journalism. Many AI-relevant indicators, for example, labor displacement or GDP growth of AI, are not reported at all or only for a few high-income countries.

For students, researchers, and policymakers who are not full-time data specialists, this makes it even harder to get a holistic view. In order to understand who benefits from AI and who bears the costs, it requires patching together economic, technical, environmental, and social datasets that live in their own silos. This fragmentation of data, combined with the speed of AI deployment and the technical complexity of both AI and sustainability, helps explain why the problem of AI-driven inequality has not been addressed in an integrated way before and why an Atlas like ours did not already exist.

III. The Solution: Atlas AI

The Website

Atlas AI has been designed as a solution to the above-mentioned problems. To tackle those we have designed the website as an interactive dashboard that provides a clear and objective view on different aspects of AI's impacts across the globe.

Atlas AI covers 34 countries, evaluated across 17 indicators, spanning across 2 years of publicly available data (2024 and 2025) within three core dimensions: environmental, social, and economic (see Figure 1). These specific dimensions were selected due to their importance and critical relevance to sustainability and inequality topics. Additionally, we implemented an interactive pop-up box that provides detailed, indicator-specific information whenever a user clicks on a region of the map.

Atlas AI was created in the following stages:

- We selected our indicators based on publicly available data.
- We collected data from trusted sources and compiled it into a single worksheet. The full dataset is available here.
- We design a friendly Human-Machine Interface (HMI) which is easy to use, interactive and compatible across devices.
- We clearly defined each indicator to help users understand its meaning and its impact on global inequality (See figure 2).
- We integrated a dynamic feature that allows users to submit requests for new indicators or contribute datasets (See figure 3).

The website's target audience consists of students who want to learn more about AI impacts, researchers and educators looking for quality data to achieve their projects.Because of its accessible, data-driven design, the Atlas AI is also targeting policymakers to help them understand the stakes of AI and providing unbiased data to support informed decision-making. The overall goal is to provide easy access to those who are aware or unaware of the costs and benefits of AI. The following section provides a brief overview of how inequality is defined within this map, followed by an explanation of the underlying dimensions and indicators used to build it.

Figure 1: An Overview of Atlas AI's Homepage

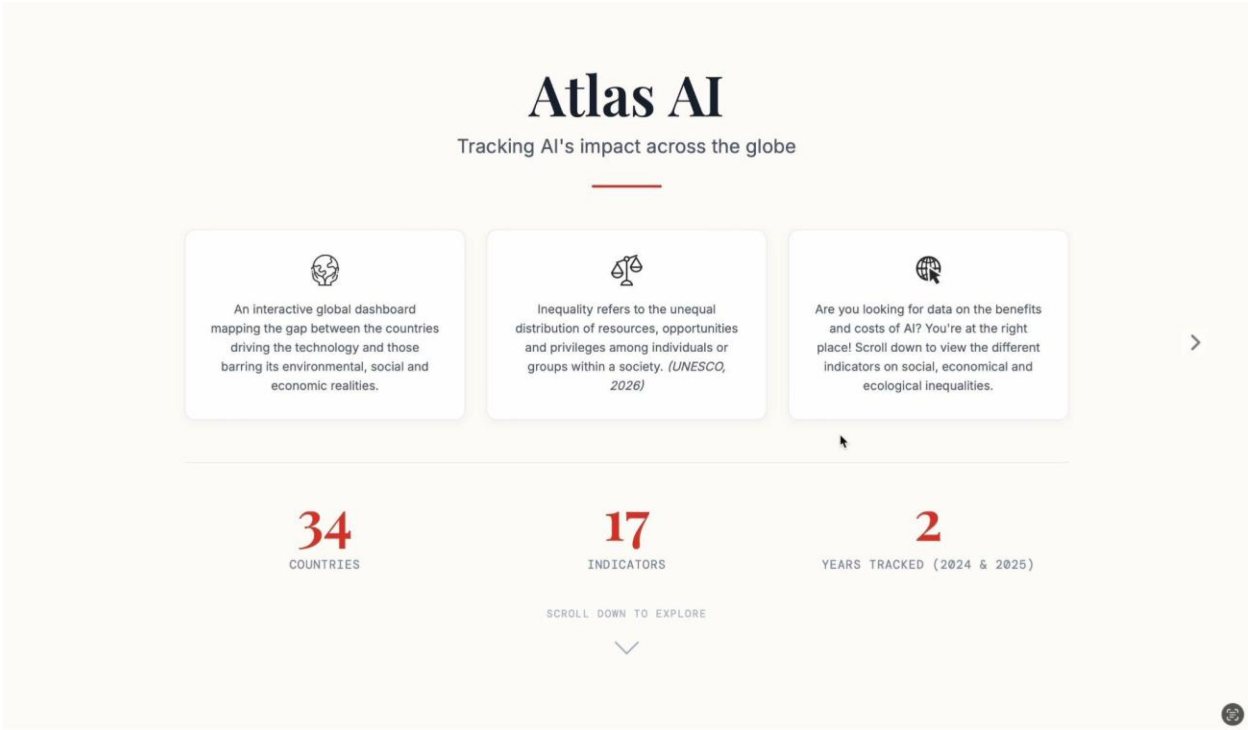


Figure 2: Dimensions and Indicators with Explanatory Pop-Ups

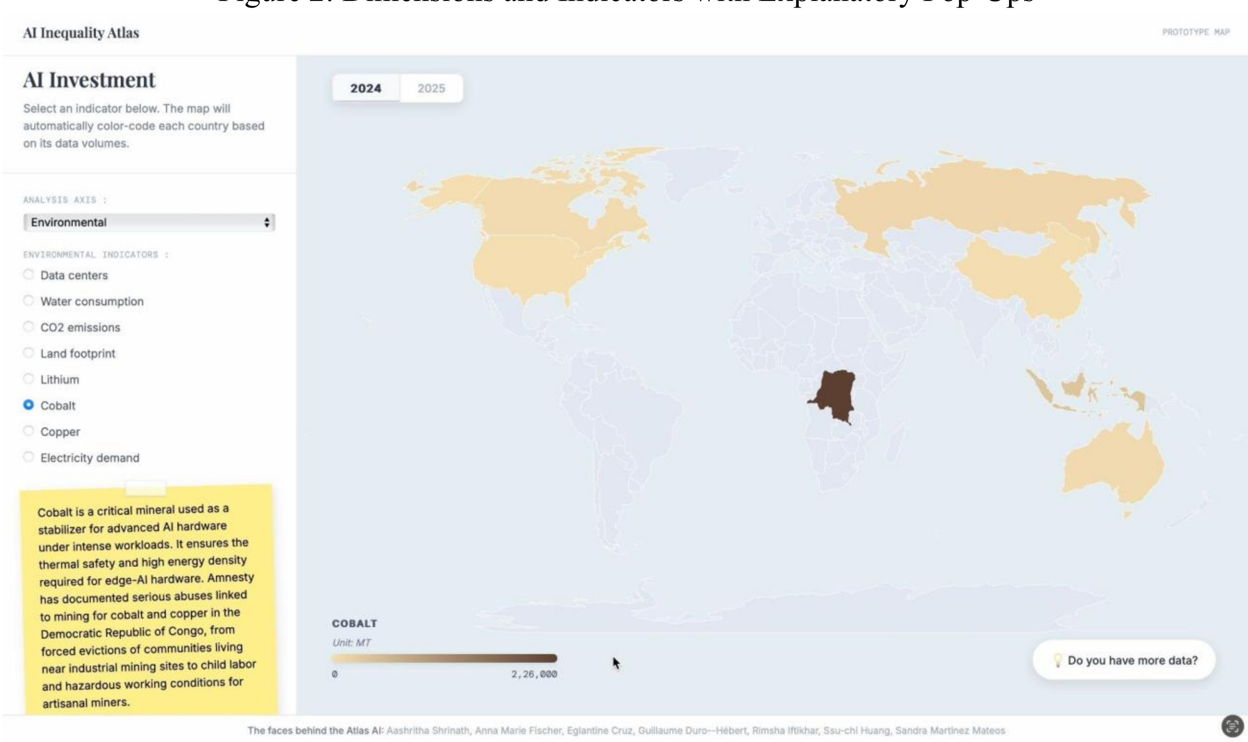
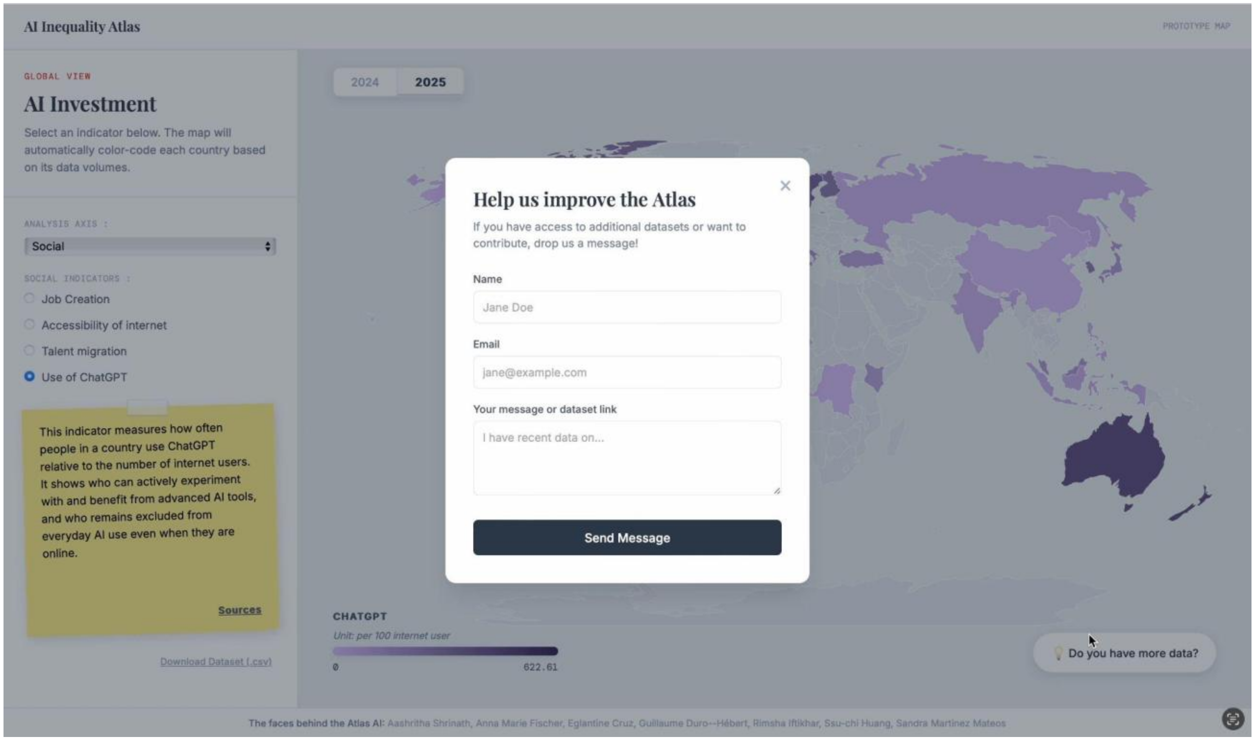


Figure 3: Dynamic Tool for Submitting New Indicators and Datasets



Definition of Inequality

UNESCO defines Inequality as the “unequal distribution of resources, opportunities, and privileges among individuals or groups within a society. It manifests in various forms, including economic disparity, gender inequality, and discrimination based on race or ethnicity.” (UNESCO, 2026)

Globally, over a billion people now use conversational AI weekly. Yet AI access and usage vary widely globally, with adoption across the global South lagging far behind the global North (IISPAI, 2026). Furthermore, there are significant differences in compute infrastructure and models between advanced economies (IISPAI, 2026). This disparity reflects, and may even reinforce, existing inequalities. AI development itself is even more concentrated: according to recent estimates, the United States of America accounts for 75% of the computing power among the world’s top 500 AI supercomputers, with China accounting for 15% (IISPAI, 2026). Companies in the United States and China also develop almost all leading general-purpose models, and a small number of countries control critical inputs for the supply chain of AI computer chips (IISPAI, 2026).
As AI technologies and large-scale data centers expand rapidly around the world, they drive increasing demands not only for minerals such as lithium, cobalt, copper and rare earth elements, but also for energy and water used to power and cool these facilities. These demands place

additional pressure on land, ecosystems and nearby communities. For instance, Amnesty International has documented, Democratic Republic of Congo, known for hosting more than 75% of the world's cobalt, serious abuses linked to mining for cobalt and copper in the Democratic Republic of Congo, from forced evictions of communities living near industrial mining sites to child labor and hazardous working conditions for artisanal miners (Amnesty International, 2026).

In Atlas AI, we map these inequalities across the same three pillars: environmental, social, and economic. Measuring inequality is inherently challenging, as its impacts vary across cultures and societies, and data transparency remains uneven and often invisible across countries and socio-economic groups. To address this, we designed specific indicators using the best available public data to provide as accurate and nuanced a picture as possible.

Analytical Dimensions & Indicators

To make these inequalities measurable, we translate each dimension (Social, Environmental, Economic) into a set of indicators that together show how AI is distributed across the globe.

Social indicators capture who is able to actively participate in the AI economy, not just access it. Jobs requiring AI skills are concentrated far more heavily in Northern countries, reflecting a broader divide in who is positioned to benefit from this technology (Pal, 2024). Internet access serves as a baseline proxy for this divide: where connectivity is limited, AI adoption is slower and its benefits less evenly shared (Cvetko, 2025). Even among those online, ChatGPT usage relative to internet users exposes a further layer of inequality. Being connected does not guarantee being able to benefit, and this gap tends to fall along the same lines as existing digital divides. ChatGPT search trends have been shown to correlate strongly with a country's human capital, meaning that even with free and open access, uptake is highest where populations are already highly skilled (Pahl, 2023). Net AI talent migration shows a similar pattern playing out among people. Countries like India train large shares of the world's AI talent, yet 67% of top AI researchers at US institutions were born abroad, showing that the countries investing in talent are often not the ones who benefit from it (Pal et al., 2026).

Environmental indicators trace the physical footprint AI leaves behind, and who is left holding it. Data centers are projected to more than double their global electricity use by 2030, largely driven by AI-related computing (European Commission, 2025). Yet, their impact goes beyond electricity: AI's water consumption could soon equal the basic annual domestic needs of 1.3 billion people, and its land footprint may reach over 14,500 square kilometers, roughly twice the size of Jakarta's metropolitan area (UN News, 2026). These costs are not evenly distributed. AI's benefits are global, but its environmental burdens concentrate in specific regions, often ones with limited capacity to absorb them (UN News, 2026). That footprint extends upstream, too. Critical minerals like lithium, cobalt, and copper are essential to AI hardware, and their extraction, particularly in the Democratic Republic of Congo, has been linked to forced evictions and child labor. Researchers increasingly describe this as an extractive supply chain in its own right, one where Global South countries supply the resources but rarely capture the value AI creates (Carayannis et al., 2025).

Economic indicators show where AI's value, ownership, and power are concentrated. Private investment in AI remains overwhelmingly concentrated in the Global North. As of September 2025, roughly 75% of new global data center capacity was being built in the United States alone (Signé, 2026). This concentration extends to who owns and produces AI knowledge. AI corporate headquarters and AI patents cluster among a small number of companies and countries. AI research publications follow the same pattern, with knowledge accumulating in the same few places rather than spreading globally. AI regulation shows how unevenly governance has kept pace, ranging from voluntary principles to binding laws like the EU AI Act.

Taken together, these three dimensions don't just describe AI's global footprint but reveal a recurring pattern: the countries generating AI's benefits and the countries absorbing its costs are rarely the same.

Social	Environmental	Economic
Jobs requiring AI skills	Number of Data Centers	Private investment in AI (in billion US $)
Internet Access	Water footprint (L per kWh)	Headquarters of AI components
Net AI talent migration	CO2 emissions (g CO2e per kWh)	AI Patents
Use of ChatGPT	Land footprint (cm2 per kWh)	AI research publications
	Critical Minerals (Lithium, Cobalt, Copper)	AI regulations
	Electricity demand	

Technical Background

Atlas AI is built using a simple and efficient technical setup that connects data to a visual map.

1. Data management

We use an SQLite database to store all the information/data we have collected. This database keeps all our data, country names, and indicators. To work with this database, we use Python with

FastAPI. The Python code acts as a messenger, when the website needs data, Python goes into the database, finds the right information, and brings it back to the map.

2. Data visualization

For what the user actually sees on the screen, we use three standard web technologies:

Technology	Function
HTML	Provides the basic structure (like the sidebar, buttons, and the map area).
CSS	Handles the styling, making the website look clean, modern, and professional.
JavaScript	This is the "active" part of the site. We use a powerful library called D3.js, which takes the data provided by our Python messenger and turns it into colors on the map. For example, when you select an indicator, JavaScript calculates how dark or light a country should be based on its data and updates the display instantly.

3. Operational Overview

When you interact with the map, like clicking on different years (2024/2025) or selecting a new indicator, the website sends a quick request to the Python back-end. Python fetches the corresponding data from the database and sends it back, and JavaScript immediately updates the colors on the map. This creates a smooth, interactive experience where data becomes visual and easy to understand.

For now, the website is hosted on Render, which is the best solution we have at the moment, but it's definitely not ideal. After 15 minutes of inactivity, the server falls asleep, and when someone tries to access the website, it takes about a minute to wake up, which is not the best experience for the user. Unfortunately, we currently don't have the resources to offer a more robust hosting solution.

For now, you can find it on the EME website and AI Young Leaders website:

- ai-inequality-atlas.df-eme.org
- ai-inequality-atlas.young-ai-leaders.org

Growing the Atlas Community: Vision and Outreach Strategy

We developed this Atlas AI with a clear purpose: to provide students and researchers with a comprehensive and streamlined framework for understanding the multi-dimensional inequalities surrounding global AI deployment. Beyond academia, our goal is to engage a broader ecosystem of stakeholders, fostering a collaborative community that drives mutual learning and shared insights.

To achieve this scalability, we are focusing on four key pillars:

- **Dynamic Data Crowdsourcing:** The Atlas AI features an integrated email tool allowing users to submit new datasets, dimensions, or indicators. Once verified, these inputs are integrated into the platform, ensuring that Atlas AI remains a continuously evolving, up-to-date resource.
- **TUM Student Network:** We aim to partner with student groups like the TUM Social AI Club, which collaborates with nonprofits on mission-driven AI projects. By using Atlas AI to map existing inequalities, they can pitch targeted initiatives to NGOs, such as those supporting families affected by cobalt mining in the Democratic Republic of Congo.
- **Institutional & NGO Partnerships:** We plan to engage civil society organizations directly. Key potential partners include the TUM Think Tank, which bridges research and policymaking and the applied AI Institute for Europe (UnternehmerTUM). These entities can leverage Atlas AI as an empirical knowledge base for responsible AI decision-making.
- **Digital Community:** To provide access to everyone interested in this topic, we will utilize LinkedIn and Instagram to reach a wider public audience. Additionally, Atlas AI will be hosted by the Young AI Leaders Munich network and the Deutsches Forum für Ethisches Maschinelles Entscheiden (EME), which aligns perfectly with their mission to empower youth in ethical AI governance.

IV. Discussion & Outlook

The Impact

Atlas AI gives the world something it has not had before: a single, openly accessible place where anyone like a policymaker, a student, and a journalist can see, in one view, how the AI revolution is landing differently depending on where you live. That visibility matters because inequality that cannot be seen cannot be challenged.

By tracking 17 indicators across 34 countries and two years, Atlas AI moves the conversation about AI from the abstract to the concrete. It shows not just that a gap exists between those who benefit from AI and those who bear its costs, but exactly where that gap is widest today, and which countries face the steepest disadvantages. When someone can click on the Democratic Republic of Congo and immediately understand that the country supplying the cobalt powering AI hardware is capturing almost none of its financial rewards, that is not just information, it is a tool for accountability.

The three-dimensional structure (economic, environmental, and social) of Atlas AI reflects a deliberate choice to treat AI's impact as inseparable from questions of climate justice, labour rights, and digital access. A tool that tracked only investment and patents would tell a misleading story. By sitting those figures alongside environmental costs, social indicators, and access gaps, Atlas AI makes visible the full ledger: what AI costs the planet, what it costs communities, and who is paying those costs without receiving the benefits in return.

The ambition is that this platform becomes a living reference updated, expanded, and used by researchers, civil society organizations, international institutions, and the communities most affected to negotiate fairer terms in the global AI economy.

Limitations

Atlas AI, as a first-generation product, represents the limitations of this development phase, and those limitations indicate clear paths forward. One significant barrier during development was a pervasive lack of data: many countries either do not collect or do not share comprehensive national data about their AI infrastructure and workforce. Several indicators would make Atlas AI far more informative if they existed in consistent, publicly available form, for example the job displacement by country and sector, AI's tangible contribution to GDP by country, AI exposure by gender, wages paid to data labelers, and the gap between where AI patents are issued and where the underlying AI was actually developed. These dimensions could not be included due to a lack of consistent, publicly available datasets for these indicators.

Other important areas like AI’s effect on access to health care, the effect of AI on agricultural output in developing countries, and the share of funding from public sector, private sector, and international institutions continue to be underdeveloped areas that could inform future iterations of Atlas AI as data become more widely available. These are not deadends, but rather a road map to how the next version of Atlas AI can advance and why a move towards overcoming national data secrecy, towards sustaining a shared investment in open, comparable data on AI at a global level, is a matter of equity.

Future Outlook

The Atlas AI is the first prototype that brings together scattered, fragmented indicators of AI, resource use, and inequality into one visual story. The next step is to turn it from a one-semester project into a living tool and community.

On the content side, Atlas AI will need regular updates as new data becomes available. As mentioned in the previous paragraphs, there are many more datasets we would like to include, but couldn't as the data is missing. Key gaps remain, for example, AI-specific GDP growth, labor conditions, or more detailed country coverage. A clear data maintenance plan is therefore essential, and how to document changes so that future students and researchers can build on our work rather than starting from scratch.

On the community side, we see Atlas AI as a starting point for broader conversations about AI and sustainability. In the short term, this means integrating the webpage into EME's website for online presence, sharing it on social media, and presenting it to student groups, research labs, and interested NGOs. In the longer term, Atlas AI could grow through collaborations with external partners who hold relevant datasets by adding countries, indicators, and stories.

By keeping the tool open, well-documented, and visible, we believe the project will go beyond a class assignment and evolve into a shared platform that continuously tracks how AI development reshapes our world and what trends are emerging.

V. Author Contributions

Anna Marie Fischer	Analytical Dimensions & Indicators
Aashritha Shrinath	Inequality Definition; Growing the Atlas Community
Rimsha Iftikhar	Impact, Limitations
Ssu-chi Huang	Barriers to a solution; Future Outlook
Eglantine Cruz	Introduction; Technical background
Guillaume Duro-hebert	Basic information about the Website
Sandra Martinez Mateos	Problem definition

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